



8.6.4 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept below.



I need help.



I got it.



I could help someone.

Properties of a parallelogram			
Properties of a rectangle			
Properties of a kite			
Completing proofs using properties and theorems about parallelograms			
Using the properties of a parallelogram to solve problems			
Properties of a trapezoid			
Using the properties of a trapezoid to solve problems			

Unit 8, Lesson 7: Trapezoid Midsegment Theorem



Benchmark

MA.912.GR.1.5 Prove relationships and theorems about trapezoids. Solve mathematical and real-world problems involving postulates, relationships, and theorems of trapezoids.



Learning Targets

- ☐ I can prove the trapezoid midsegment theorem.
- ☐ I can use properties of trapezoids to solve mathematical and real-world problems.



8.7.1 Warm-Up: Trapezoids in Architecture

Four different buildings are shown.

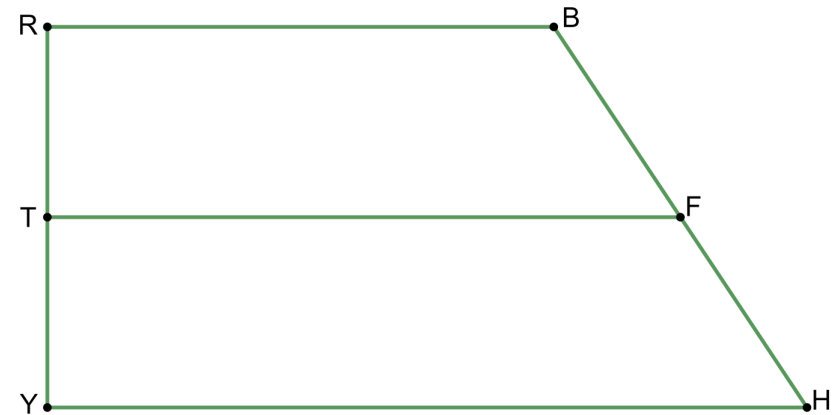
Machu Picchu, Peru	Greenland Center, Beijing, China
	
Garden Terrace, Nagasaki	International Spy Museum, Washington D.C.
	

1. What do you notice about the buildings?
2. Describe how a trapezoid is used in each building.



8.7.2 Exploration: Midsegment of a Trapezoid

1. Trapezoid $RBHY$ is shown, where $\overline{RB} \parallel \overline{YH}$.



- a. Use patty paper or a ruler to determine if $\overline{TF}$ is the midsegment of trapezoid $RBHY$. Justify your conclusion.
- b. Use a ruler to complete the table.

Segment(s)	Measure
$\overline{RB}$	
$\overline{YH}$	
$\overline{TF}$	

2. What do you notice about the relationship between the length of the midsegment and the sum of the lengths of the bases?

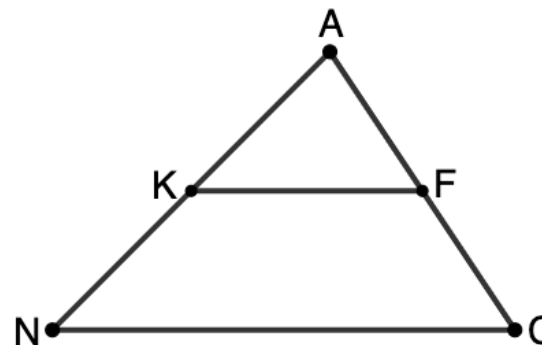
3. Write an equation to show the relationship between the length of the midsegment of a trapezoid and the length of its bases.



8.7.3 Guided Instruction: Trapezoid Midsegment Theorem

The **midsegment** of a triangle is a segment joining the midpoints of two sides of a triangle.

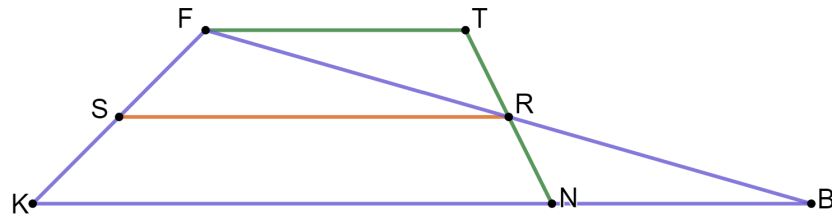
1. $\triangle ACN$ is shown with midsegment $\overline{KF}$.



Fill in the blanks to complete the triangle midsegment theorem.

The triangle midsegment theorem states that the midsegment of a triangle is _____ to the third side it does not touch and is _____ the length of the third side.

2. $\triangle FBK$ with midsegment $\overline{SR}$ and trapezoid $FTNK$ are shown where $\overline{FT} \parallel \overline{KN}$.



a. Explain why $SF = KS$.

b. If $\overline{NB} \cong \overline{FT}$, justify why $\overline{NR} \cong \overline{RT}$.

c. Discuss with your partner what information is necessary so that the definition of a trapezoid can be used to state that $FTRS$ is a trapezoid. Summarize your discussion.

- d. Discuss with your partner what else needs to be determined so that it can be stated that $\overline{SR}$ is a midsegment of $FTNK$. Summarize your discussion.



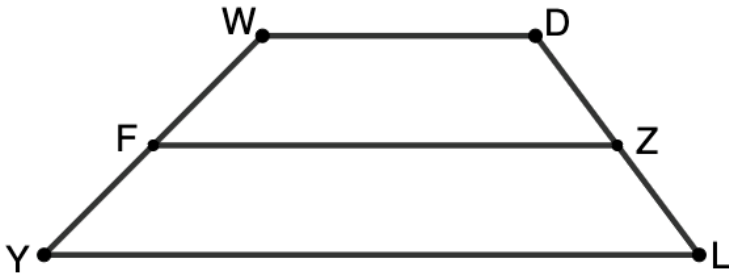
Trapezoid Midsegment Theorem

The midsegment of a trapezoid is parallel to the bases and is half the sum of the bases.



8.7.4 Guided Practice: Using the Trapezoid Midsegment Theorem

1. Trapezoid $WDLY$ with midsegment $\overline{FZ}$ is shown.

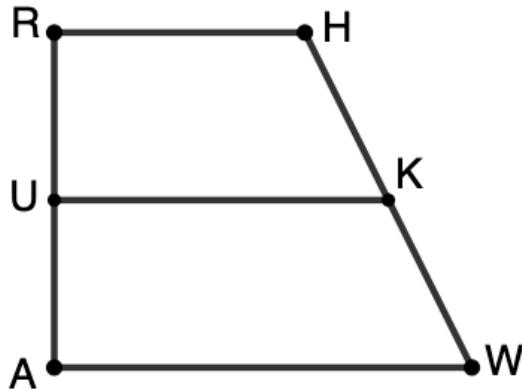


- a. If $WD = 14.2$ and $YL = 29.7$, what is the length of $\overline{FZ}$?
- b. If $WDLY$ is an isosceles trapezoid and $m\angle DLY = 53^\circ$, what is $m\angle WFZ$?
- c. If $FZ = 11.4$ and $YL = 17.6$, what is the length of $\overline{WD}$?
- d. What is the value of k if $WD = 4k + 4.2$, $FZ = 7k + 2.1$, and $YL = 6k + 16.8$?
- e. If $WF = \frac{2}{3}p$ and $WY = 12 - 2p$, what is the length of $\overline{WY}$?

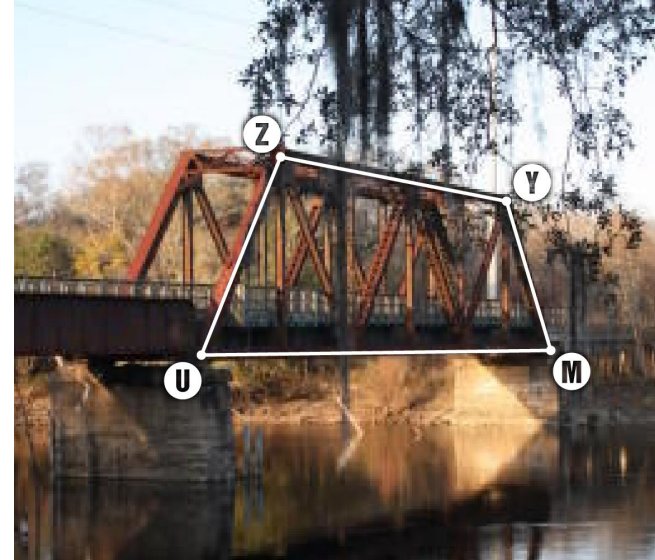


8.7.5 Your Turn!

1. Trapezoid $RHWA$ is shown with midsegment $\overline{UK}$. Find the values of x and y if $RU = -2x - 2y$, $AU = 10$, $RH = 4x$, $UK = 14$, and $AW = -y + 8$.



2. Located on the Nature Coast State Trail in north-central Florida is the Suwannee River Trestle, a bridge that is shaped like an isosceles trapezoid. The state has been inspecting older bridges to ensure their safety and has determined that the Suwannee River Trestle, represented as trapezoid $ZYMU$, needs an additional support brace through its middle.



The support brace is to connect sides $\overline{ZU}$ and $\overline{YM}$ at their midpoints. If the length of leg $ZU = 28$ feet (ft) and the perimeter of the trestle measures 153 ft, what is the length, in ft, of the new support brace?



8.7.6 Wrap-Up: Cool Down

1. Your friend is absent today and will need the notes from class. Write a quick note to help them understand the trapezoid midsegment theorem.

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